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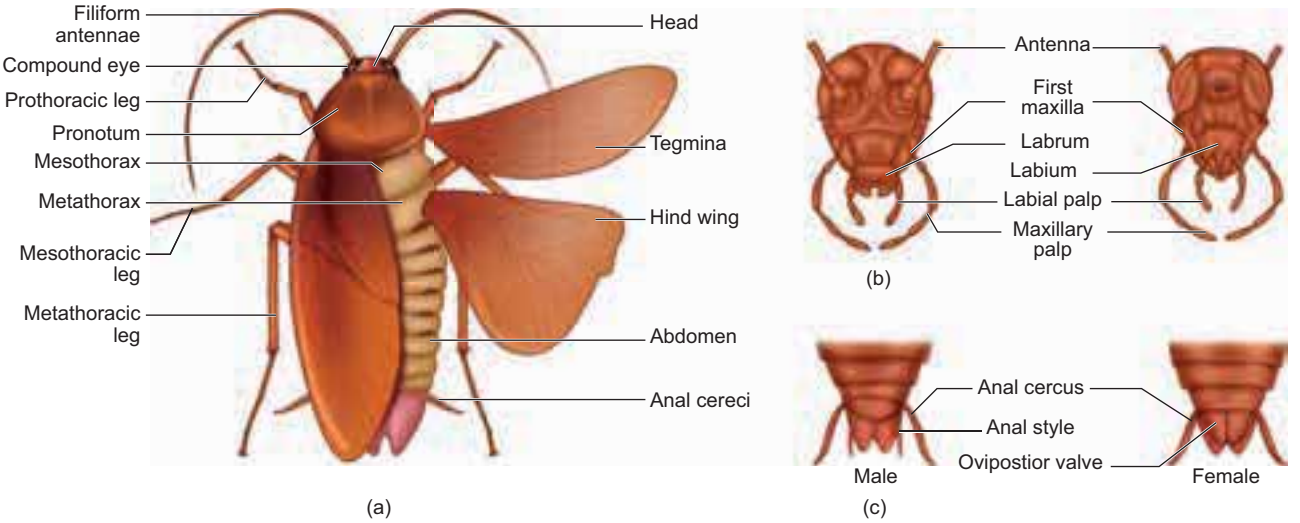


Figure 4.8 *Periplaneta americana:* (a) External features (b) Head dorsal and ventral view (c) Male and Female ventral view of posterior segment of abdomen

by a hard, brown coloured, chitinous exoskeleton. In each segment, exoskeleton has hardened plates called **sclerites**, which are joined together by a delicate and elastic **articular membrane** or **arthrodial membrane**. The sclerites of the dorsal side are called **tergites**, those on the ventral side are called **sternites** and those of lateral sides are called **pleurites**.

The head of cockroach is small, triangular lies at right angle to the longitudinal body **axis**. the mouth parts are directed downwards so it is **hypognathous.** It is formed by the fusion of six segments and shows great mobility in all directions due to a flexible neck (Figure 4.8). The head capsule bears a pair of large, sessile, and reniform **compound eyes**, a pair of antennae and appendages around the mouth. Antennae have sensory receptors that help in monitoring the environment. The appendages form the mouth parts which are of biting and chewing type (**Mandibulate** or **Orthopterus type**). The mouth parts consist of a **labrum** (upper lip), a pair of **mandibles**, a pair of **maxillae**, a **labium** (lower lip) and a **hypopharynx** (tongue) or **lingua** (Figure 4.9). The thorax consists of three segments – **Prothorax**, **Mesothorax** and **Metathorax**. The prothoracic segment is the largest. The head is connected with thorax by a short extension of the prothorax called as the **neck** or **cervicum**. Each thoracic segment bears a pair of walking legs. Due to the presence of three pairs of walking legs it is also called **hexapoda** (hexa-six, poda-feet) All the three pairs of walking legs are similar and each leg consists of five segments – **coxa** (large), **trochanter** (small), **femur** (long and broad), **tibia** (long and thick) and **tarsus**. The last segment of the leg - tarsus has five movable joints or **podomeres** or **tarsomeres**. Cockroach has two pairs of wings, the first pair arises from mesothorax and protects

The cockroaches are ancient and most basic among all groups of insects, dating back to the carboniferous period, about 320 million years ago.